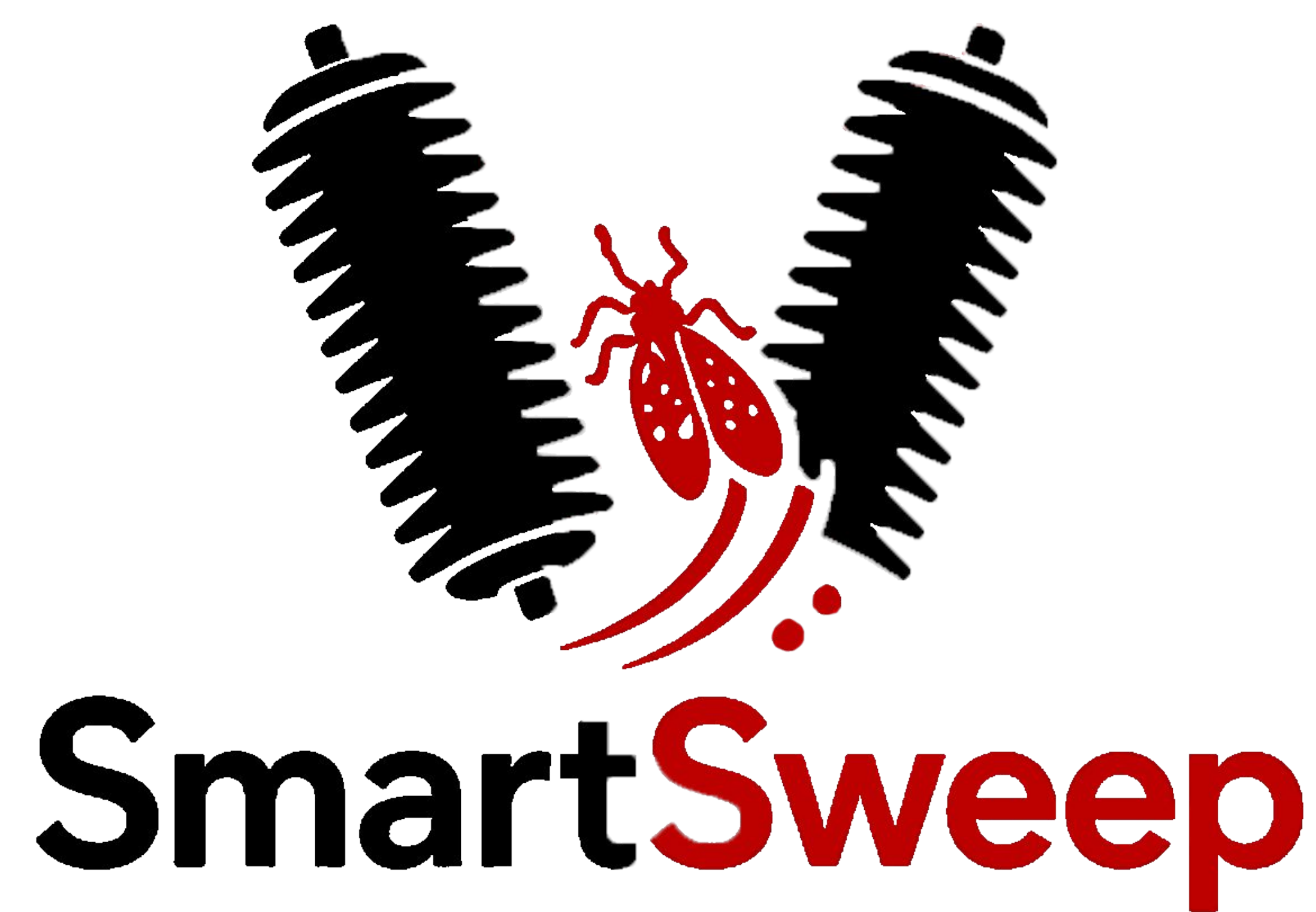


Spotted and Stopped

Removing Spotted Lanternflies From Vines Before Harvesting



Subproblem: Harvests Contamination from SLF

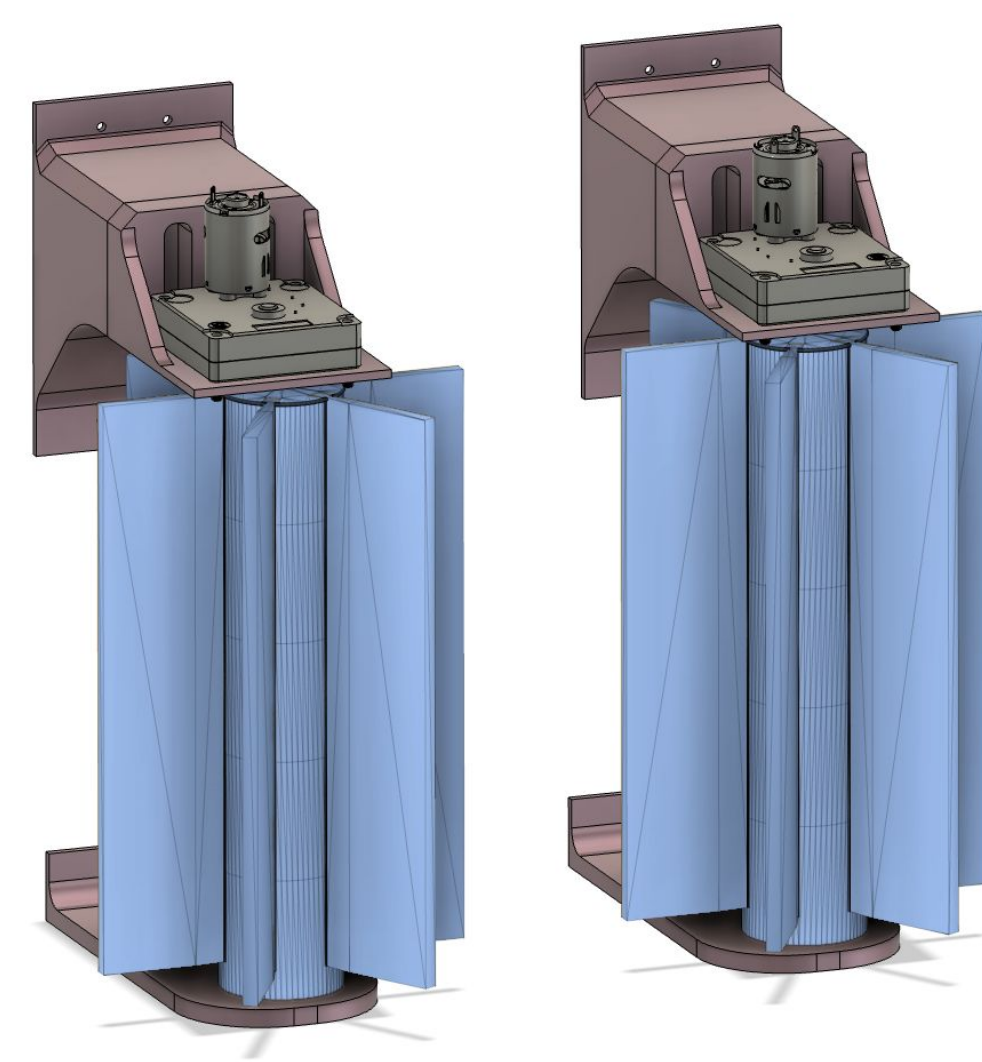
- Farmers invest vast amounts of time and resources into each crop, all for it to be ruined in the harvesting process
- In New York, operators primarily use mechanical harvesting methods, making it difficult to separate insects from grapes
- A Cornell-Penn State study found 60% of spotted lantern flies on vines end up in harvester bins
- Rejection if 1-2 adult insects found per 1kg sample, by regulation
- Reducing contamination increases yield and farmer profit

Solution Direction: Motorized brush module for SLF Removal

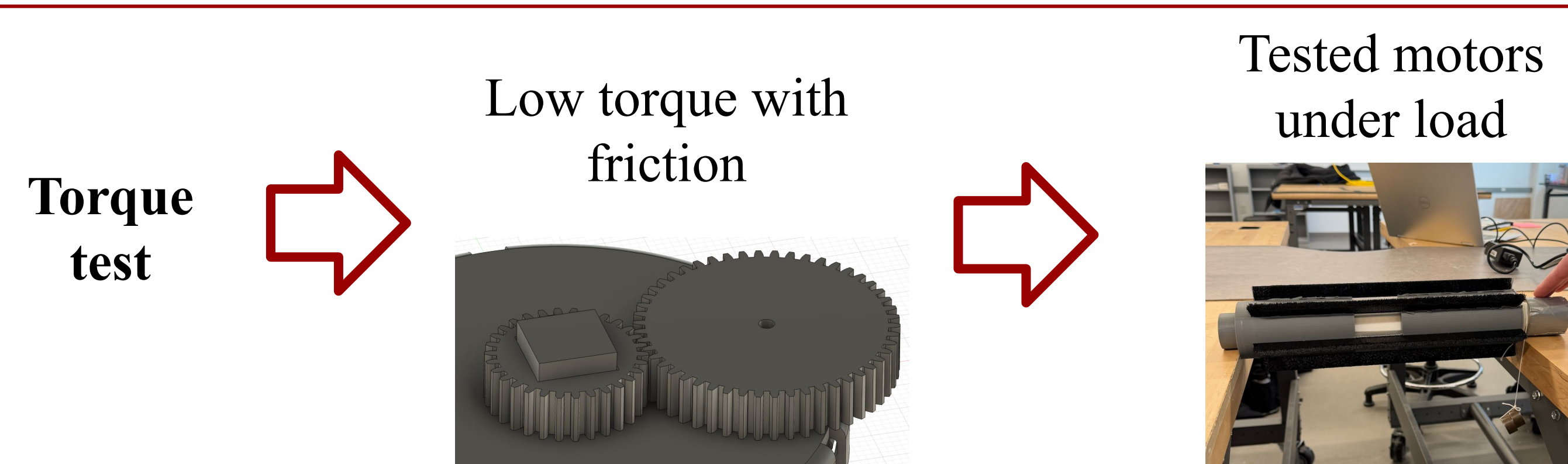
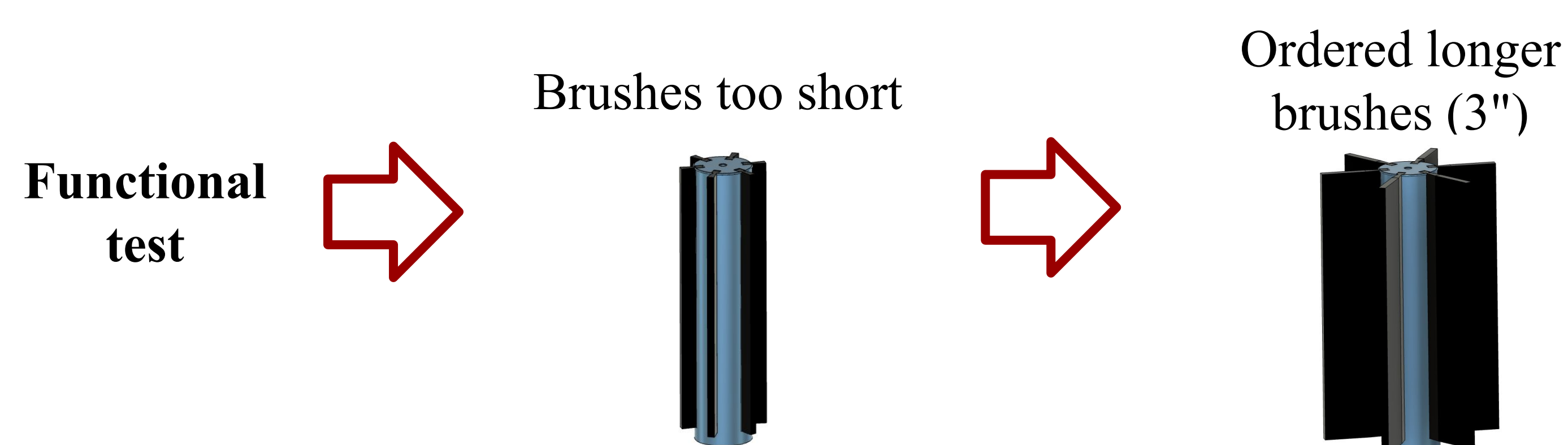
Opposing rotating brushes dislodge SLF and push them away from machine just before harvest.

Key benefits of this design include:

- Compatible with existing harvesters
- Mounts to multiple models (minor mods)
- Adjustable to different vine sizes
- Eliminates post-harvest filtering



Initial Testing: Design Improvements



Final Prototype: Harvester-Mounted Brush Module

Key features of our prototype:

- Shaft with slots for 6 replaceable strip brushes
- 12V DC motor (47 rpm, 160 in*ozf torque)
- Ball bearing for smooth rotation
- Top bracket supports motor and wiring
- Bottom bracket supports shaft and integrates bearing for smooth rotation

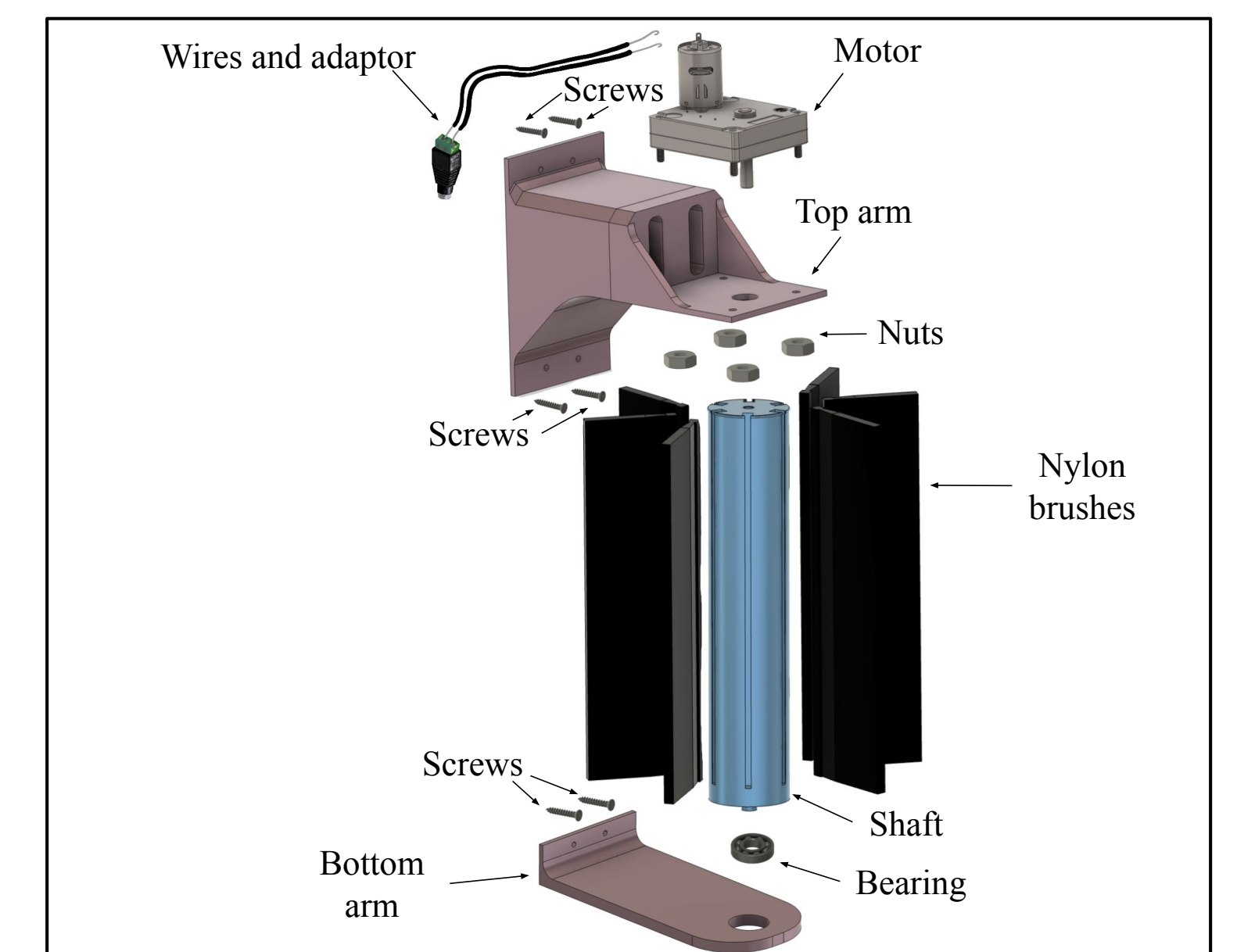


Fig. 1: Exploded view of final prototype, showing all CAD-designed components for assembly

Tests & Results: Outward Rotations Maximizes Removal ~ 70% Effective on Paper Clips

Test 1: Functional Test

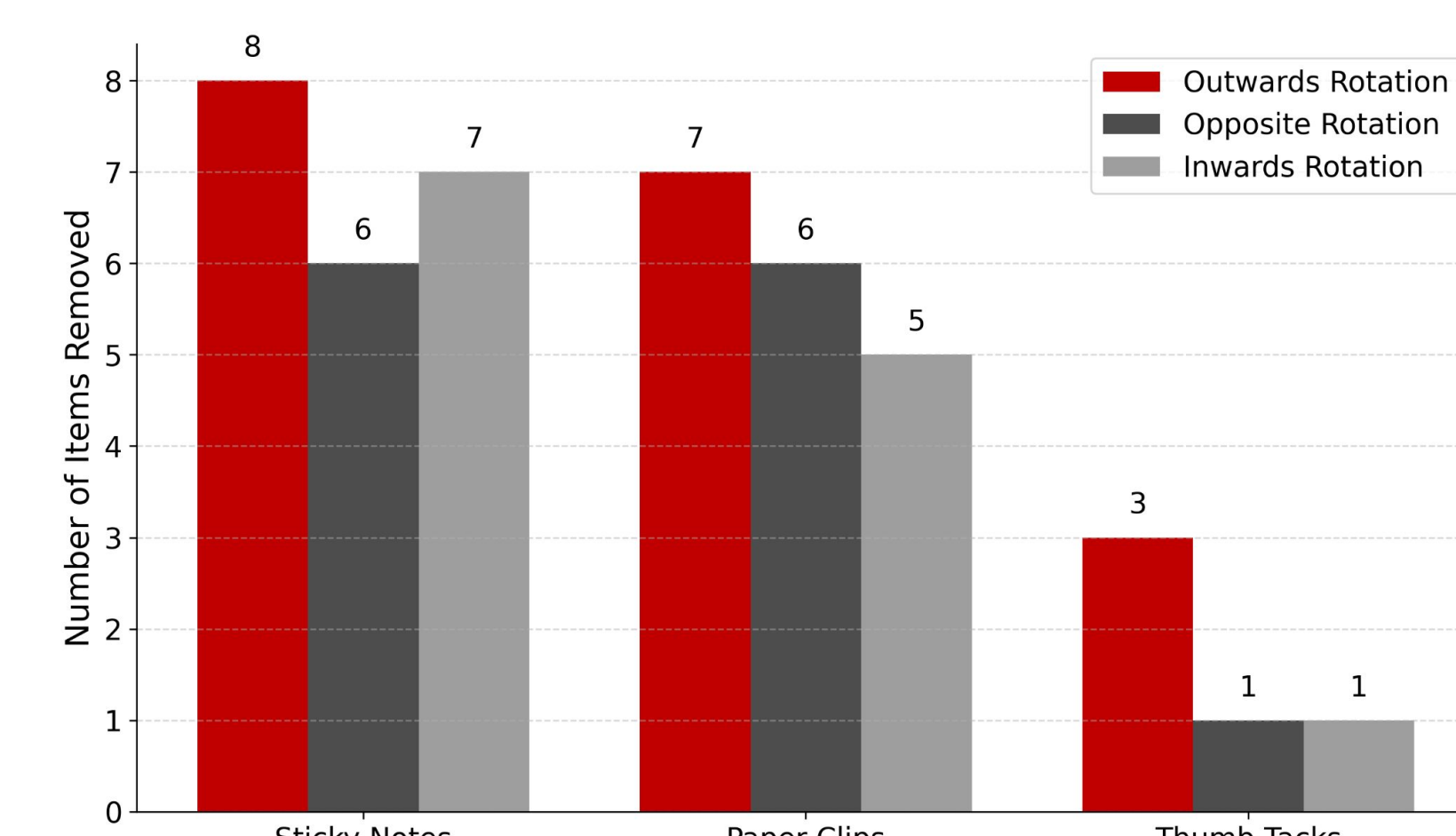


Fig. 2: Number of items removed for different resistance types and rotation directions. A cylindrical tube containing 10 items of each type was passed through the brushes. Outward rotation removed the most items, with thumb tacks representing the upper limit of effective removal.

Test 2: Assembly Time Evaluation

Task	Test 1 (s)	Test 2 (s)	Test 3 (s)	Avg (s)
Replacing Brushes	42	37	44	41
Mounting on Bracket	321	247	268	279

Table 1: Time Required for Key Assembly Tasks. Replacing brushes takes under 1 minute on average, whereas mounting the assembly on the bracket takes significantly longer (~279 s), making it approximately 6.8× more time-consuming and the dominant contributor to assembly time.

Test 3: Effect on Exposed Grapes

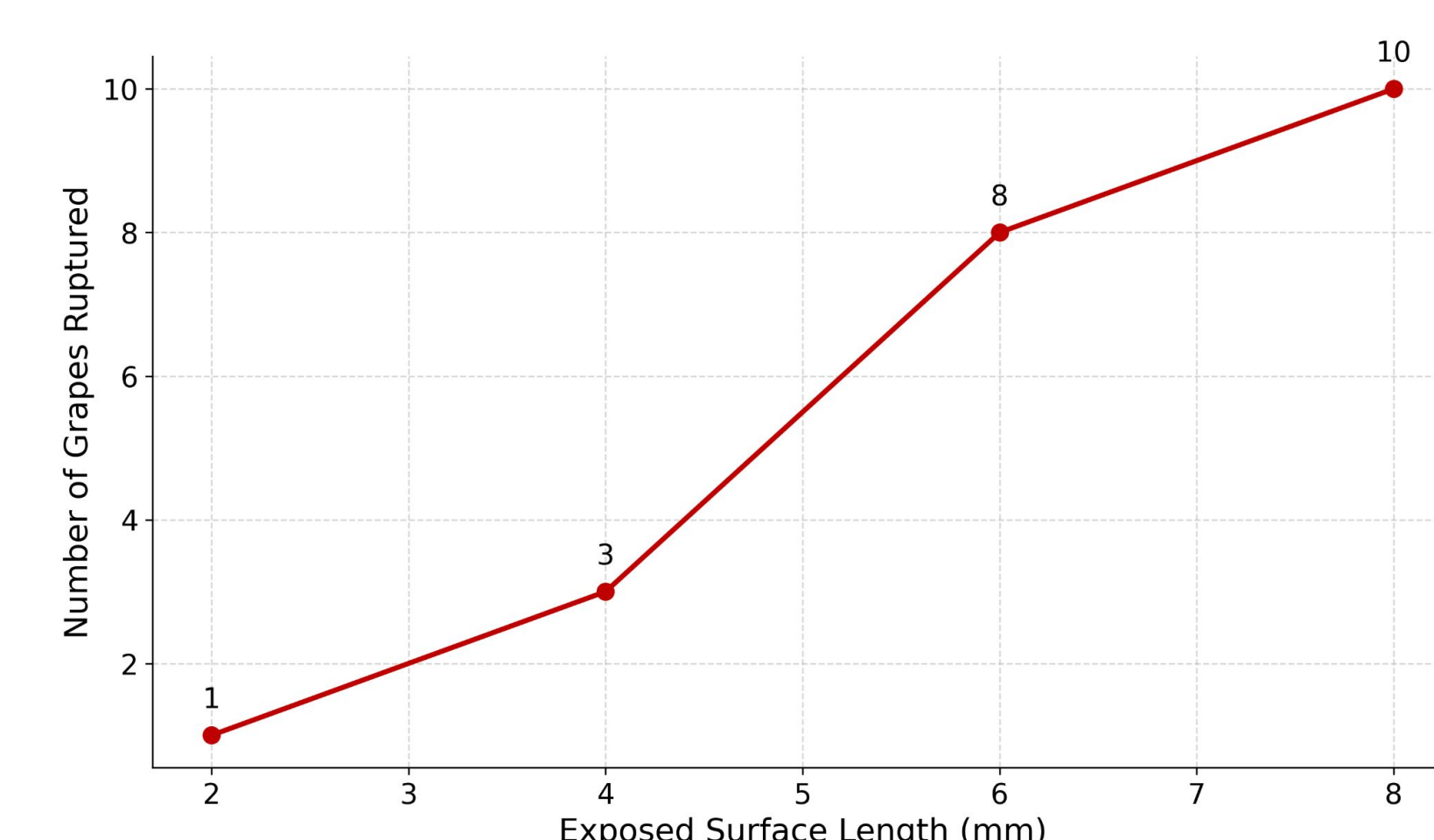


Fig. 3: Grapes ruptured vs. exposed surface length (10 grapes tested). Exposed lengths of more than 6mm result in losses of 80%.

Conclusion & Recommendations: Promising, Needs Refinement

Success Criteria Evaluation

- Passed S.C. of removing 80% of sticky notes (functional test)
- Passed S.C. for brush replacement time < 1 min
- Passed S.C. for brushes remaining secure during operation

Recommendations and Next Steps

- Increase motor power (higher torque & RPM)
- Simplify and standardize mounting on different harvesters
- Optimize brush design (stiffer bristles, controlled exposure length)
- Improve shaft stability to reduce vibration
- Minimize grape damage by limiting brush contact (< 6mm exposure)

